MAHAM FAIZ MALIK

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EDUCATION

MS Energy Systems Engineering Sept 2025 - Present
NUST - National University of Sciences and Technology, H-12 Islamabad

Courses related to Energy Systems and Technologies: Materials Science and Engineering, Applied Solar Energy, Wind Energy, Fuel Cells, Energy Resources and Technologies

Course related to Modeling, Simulation and Systems Analysis: Modeling of Energy Systems

Courses related to Climate, Environment and Sustainability: Energy and Climate Change, Energy and Environment

CGPA: 3.73/4.00

BS Physics Sept 2021 - July 2025
PIEAS - Pakistan Institute of Engineering and Applied Sciences, Islamabad

Courses related to Physics: Mathematical Methods of Physics, Quantum Mechanics, Classical Mechanics, Electromagnetic Theory, Electronic Circuit Design, Electronic Devices and Circuits, Thermal and Statistical Physics, Modern Physics, Optics, Heat and Thermodynamics, Basic Nuclear Engineering, Particle Physics, Quantum Field Theory

Courses related to Mathematics: Calculus, Linear Algebra, Mathematical Methods of Physics, Probability and Statistics, Ordinary Differential Equations

Courses related to Computer Science: Computer Fundamentals and Programming, Intro to Machine Learning

CGPA: 3.50/4.00

TECHNICAL & SOFT SKILLS

Programming & Development: Python, C, MATLAB
Data Analytics & Visualization: Power BI, Advanced Excel (Pivot Tables, Dashboards, Data Analysis), Python(Pandas, NumPy, Matplotlib)
Energy Modeling & Simulation Tools: RETScreen Expert, PVsyst, SAM (System Advisor Model)
Machine Learning & Scientific Computing: Machine Learning, TensorFlow, Cirq, Lyapunov
Documentation & Reporting Tools: LaTeX, Microsoft Word, Microsoft PowerPoint
Languages: English, fluent

WORK EXPERIENCE

Intern (PTPRI - Pakistan Tokamak Plasma Research Institute) July 2023 - August 2023
Project : Analytical Analysis of Transient Heat Transfer
- Modeled transient heat transfer in plasma gasification chamber systems and comparison with ANSYS.
- Learned plasma physics fundamentals and reactor material. behavior.

Intern (NILOP - National Institute of Laser and Optronics) July 2024 - August 2024
Introductory Quantum lab experiments
- Performed quantum optics and laser experiments.
- Conducted single-photon and optical detection measurements.

Quantum Workshop (PIEAS) February 17 - 21, 2025

• Participation in international workshop on Recent Advances in Quantum Science and Technology.

PROJECTS

Wind Farm Design and Energy Analytics Project – 200 MW Chagai Wind Farm 2026

• Collected, validated, processed, and analyzed large wind resource datasets using Python (Pandas, NumPy) and visualization tools. Performed wind-speed distribution analysis and energy yield estimation for a 200 MW wind farm. Applied simulation tools for micrositing optimization, wake-loss assessment, and production forecasting.

• Developed data-driven insights supporting turbine placement, grid integration, and project feasibility studies.

Techno-Economic Analysis of Residential Solar Water Heating Systems and Residential Off-Grid Solar PV System January 2026

• Analyzed energy-efficiency and performance indicators, conducted sensitivity analysis, and generated data-driven recommendations of residential solar water heaters in RET--Screen Expert, comparing evacuated tube and glazed collectors to assess financial viability and potential for replacing gas-based water heating and optimizing collector slope for maximum efficiency.

• Modeled an off-grid residential solar PV system in RETScreen Expert, to evaluate feasibility of shifting household electricity load from grid power to solar energy.

Stand-Alone PV Water-Pumping System November 2025

• Completed the full design of a standalone photovoltaic water-pumping system, performing solar irradiance assessment, hydraulic load analysis, PV array sizing, component selection, performance simulations for off-grid operation and analyzed energy-efficiency, conducted sensitivity analysis, and generated data-driven recommendations using PVsyst.

Entanglement in Coupled Cavity Optomechanics (Final Year Project) Sept 2024 - July 2025

• Conducted an in-depth study of optomechanical systems, analyzing how energy variations affect the system Hamiltonian. Focused on entanglement in tripartite optomechanical systems via optical-dark-mode control, investigating how breaking the optical dark mode enhances entanglement. Reproduced and validated all results from the research paper*Tripartite Optomechanical Entanglement via Optical-Dark-Mode Control* (Deng-Gao Lai et al., RIKEN & Univ. of Michigan, 2023), theoretical models and simulations using Python Lyapynov libraries.

Quantum-Assisted Binary Image Classification (Machine Learning Project) 2025

• Implemented and evaluated a quantum-assisted binary image classification model using TensorFlow Quantum and Cirq, improving accuracy through hybrid quantum-classical circuits and gaining hands-on experience with quantum machine learning and parameterized quantum circuit design.

Data Analysis Project (Semester Project) 2023

• Survey Analysis of Anxiety Levels in University Students. Performed statistical analysis with Python.

Seat Allocation System for Pakistan Eagle Flight PE-49 and ATM Machine Implementation Using C Language (Semester Projects) 2022

• Development of software using C language to implement a seat allocation system for Pakistan Eagle Flight PE-49, handling seat selection, fare calculation, and displaying messages based on seat availability and class type.

• Development of a C language code for the implementation of ATM machine.

CERTIFICATES AND ACHIEVEMENTS

Best Final Year Project Certificate in Bachelors (ORIC - PIEAS) 2025

Certificate of Appreciation for winning Cricket and Tug of War in Sports Week (PIEAS) 2023